KEVIN REN

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Education

Princeton University, Princeton, NJ — **September 2022 - present**

- PhD in Mathematics, advised by Prof. Assaf Naor — Expected 2027
- Master of Science in Mathematics — October 2023

Massachusetts Institute of Technology (MIT), Cambridge, MA

- Bachelor of Science in Mathematics and Physics. GPA: 5.0/5.0 — **February 2022**

Skills

Programming Languages: C++, Python, Java, Mathematica, SageMath, MatLab

Languages: Chinese (fluent), English (native)

Research and Work Experience

Princeton Graduate Program in Mathematics — **September 2022-present**

- Worked on cutting-edge research problems in analysis, geometry, and algorithms.
- Invited as a seminar speaker by MIT, Princeton, Berkeley, UPenn, NYU, and other universities.
- Produced a dozen papers for publication, of which two were featured in *Quanta* magazine, and one was featured in the 2025 Symposium on Theory of Computing (STOC) conference.

Jane Street Internship — **Summer 2024**

- Used financial principles and data analysis in Python to identify signals in financial data.
- Conducted hypothesis testing and statistical feasibility analysis on identified signals.
- Engaged in frequent team-based mock trading sessions in fast-paced environments.

Teaching Experience

Teaching Assistant for Princeton MAT 202 (Linear Algebra)	**Spring 2025**
Teaching Assistant for Princeton MAT 103 (Calculus I)	**Fall 2024**
Undergraduate Assistant for MIT 18.103 (Fourier Analysis)	**Fall 2020**
Grader for Math Olympiad Program (MOP)	**Summer 2021**
Instructor for San Diego Math Circle	**2016-2021**

Honors and Awards

Cubist Point72 PhD Fellowship	**2025-26**
Simons Dissertation Fellowship in Mathematics	**2025-27**
Top 70 in Alibaba Global Math Competition	**2021-23**
NSF GRFP Fellowship	**2022**
Top 25 in Putnam Competition	**2021**

Siemens Competition National Finalist	**2017**
USA Computing Olympiad Platinum Division qualifier	**2017**
USA Math Olympiad Winner, Akamai Award for 3rd place	**2015, 2016**

List of Research Papers

[1] Incidence bounds related to circular Furstenberg sets (joint with John Green, Terence L. J. Harris, Yumeng Ou, Sarah Tammen). *arXiv preprint*, 2025.

[2] Euclidean embedding, randomized clustering, and Lipschitz extension for finite and doubling subsets of L_p when $p > 2$ (joint with Assaf Naor). *arXiv preprint*, 2025.

[3] Random zero sets with local growth guarantees (joint with Alan Chang, Assaf Naor). *STOC '25: Proceedings of the 57th Annual ACM Symposium on Theory of Computing.*

[4] On the projections of almost Ahlfors regular sets (joint with Tuomas Orponen). *arXiv preprint*, 2024.

[5] Radial Projections in $\mathbb{R}^n$ Revisited (joint with Paige Bright and Yuqiu Fu). *arXiv preprint*, 2024.

[6] Sobolev extension in a simple case (joint with Marjorie Drake, Charles Fefferman, and Anna Skorobogatova). *Advanced Nonlinear Studies* 25 (2025), no. 2, pp. 260-278.

[7] Weighted refined decoupling estimates and application to Falconer distance set problem (joint with Xiumin Du, Yumeng Ou, and Ruixiang Zhang). *arXiv preprint*, 2023.

[8] New improvement to Falconer distance set problem in higher dimensions (joint with Xiumin Du, Yumeng Ou, and Ruixiang Zhang). *arXiv preprint*, 2023.

[9] Discretized Radial Projections in $\mathbb{R}^d$. *arXiv preprint*, 2023.

[10] Furstenberg sets estimate in the plane (joint with Hong Wang). *arXiv preprint*, 2023.

[11] A note on maximal operators for the Schrödinger equation on the torus (joint with Yuqiu Fu, Haoyu Wang). *arXiv preprint*, 2023.

[12] Incidence estimates for alpha-dimensional tubes and beta-dimensional balls in the plane (joint with Yuqiu Fu). *J Fractal Geometry* 11 (2024), no. 1/2, pp. 1–30.

[13] An incidence estimate and a Furstenberg type estimate for tubes in the plane (joint with Yuqiu Fu and Shengwen Gan). *J Fourier Anal Appl* 28, 59 (2022).

[14] %-Immanants and Temperley-Lieb Immanants (joint with Frank Lu, Dawei Shen, Siki Wang). *arXiv preprint*, 2023.

[15] Topology of Augmented Bergman Complexes (joint with Elisabeth Bullock, Aidan Kelley, Victor Reiner, Gahl Shemy, Dawei Shen, Brian Sun, Amy Tao, Zhichun Joy Zhang). *Electronic Journal of Combinatorics*, p. 1-31 (2022).

[16] Extending the Graph Formalism to Higher-Order Gates (joint with Andrey Boris Khesin). *Quantum Inf. Comput.* 23 (2023), no. 13-14, 1128–1141.

[17] A flexible ChIP-sequencing simulation toolkit (joint with An Zheng, Michael Lamkin, Yutong Qiu, Alon Goren, Melissa Gymrek). *BMC bioinformatics* 22 (2021), 1-10.

List of Invited Talks

Brin MRC Workshop on Sparse Equidistribution	**September 2025**
AIM Workshop on Metric Embeddings	**July 2025**
ACM Symposium on Theory of Computing	**June 2025**
MIT Analysis Seminar	**April 2025**
NYU Analysis Seminar	**February 2025**

Princeton Analysis Seminar	**February 2025**
University of North Carolina Analysis Seminar	**October 2024**
Northwestern Analysis Seminar	**April 2024**
AMS Spring Central Sectional Meeting	**April 2024**
UCLA Department Colloquium	**April 2024**
UPenn Analysis Seminar	**February 2024**
Workshop in Analysis at Georgia Institute of Technology	**December 2023**
Berkeley Analysis Seminar	**October 2023**
University of British Columbia Analysis Seminar	**October 2023**
Online Analysis Research Seminar	**September 2023**
Harmonic Analysis and Fractal Sets conference at Ohio State University	**March 2023**
Princeton Graduate Student Seminar	**October 2022**
Joint Mathematics Meetings AMS/PME student poster session	**January 2022**

Articles about my research

Leila Sloman. Number of Distances Separating Points Has a New Bound. https://www.quantamagazine.org/number-of-distances-separating-points-has-a-new-bound-20240409/

Leila Sloman. Mathematicians Cross the Line to Get to the Point. https://www.quantamagazine.org/mathematicians-cross-the-line-to-get-to-the-point-20230925/